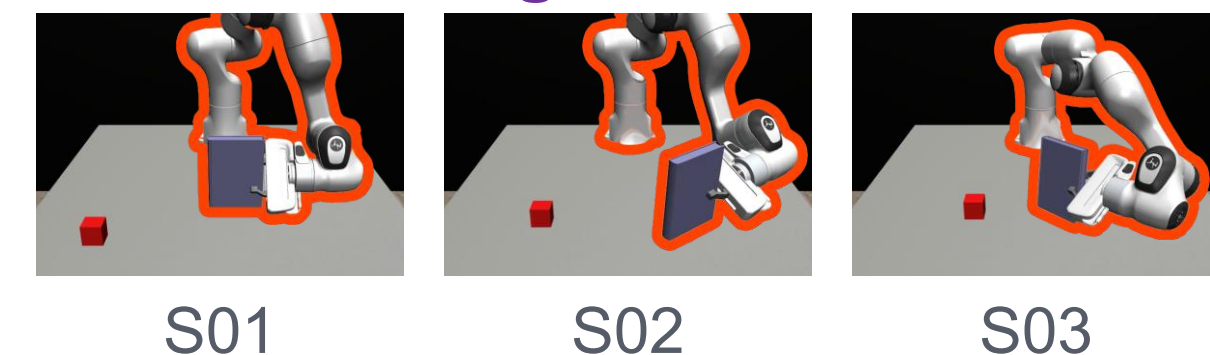


(a) Simulation Benchmarks

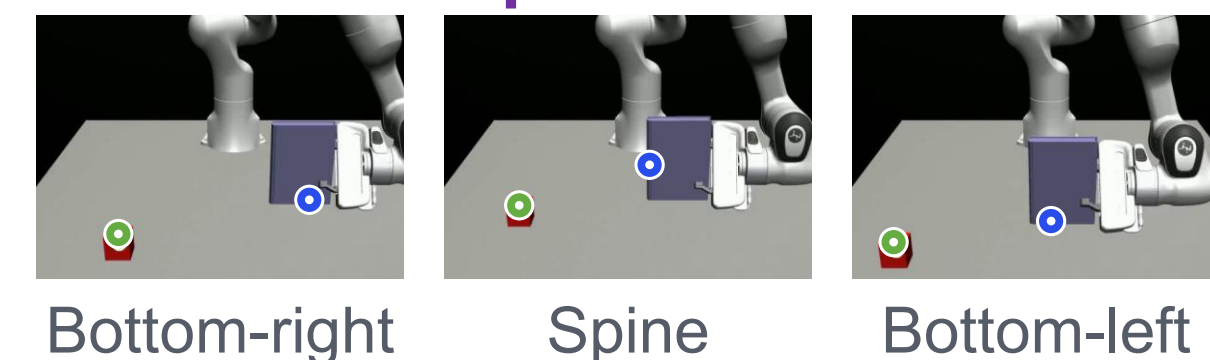
10 tools



3 initial settings



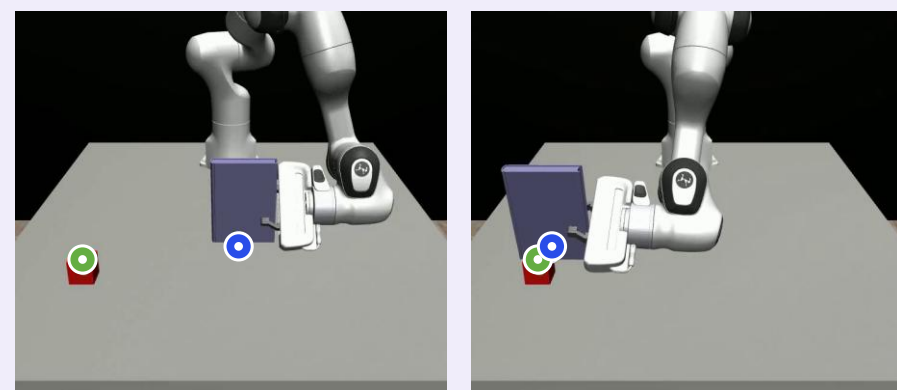
Randomized points



3 Tasks: Hit / Sweep / Hook with a Functional Point

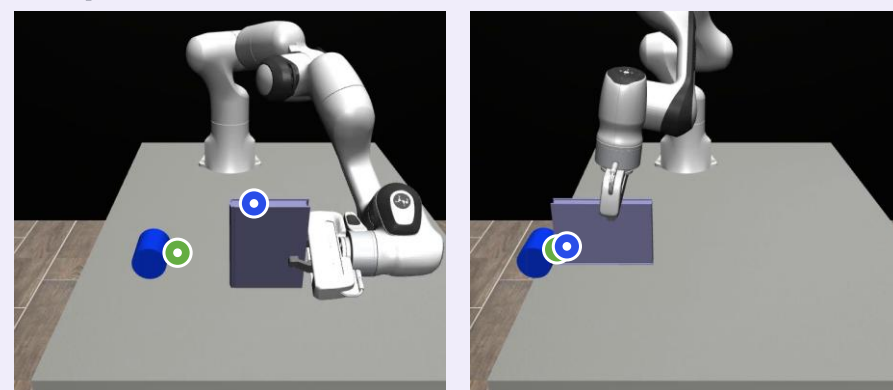
Hitting

Bottom-left



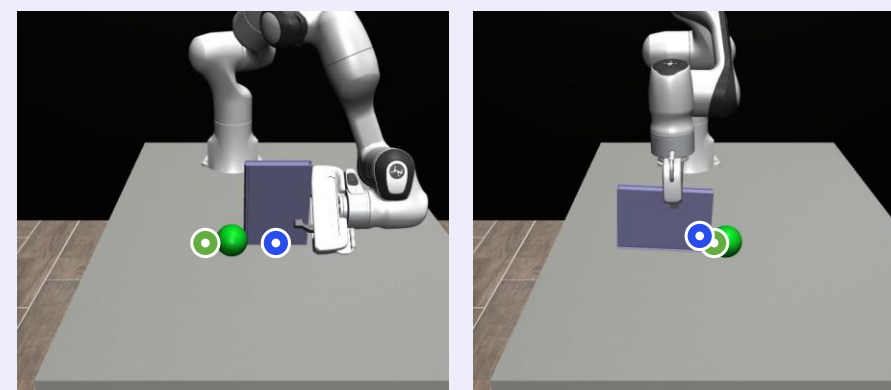
Sweeping

Top-left

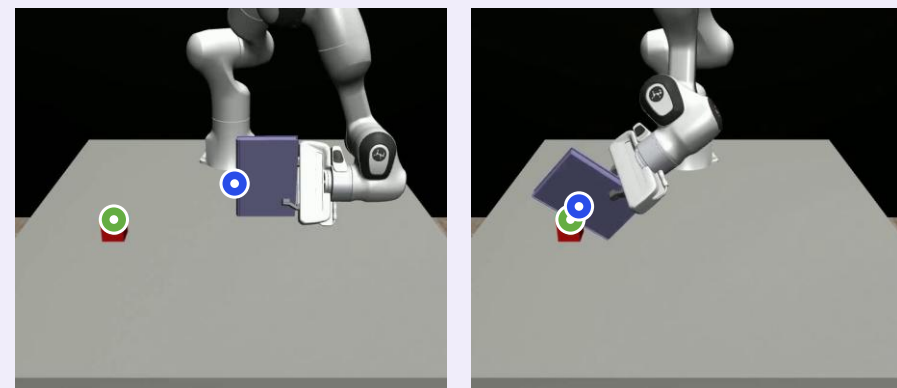


Hooking

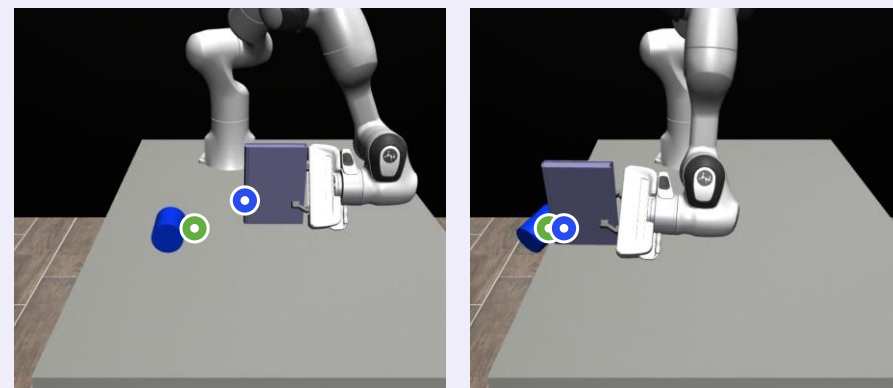
Bottom-left



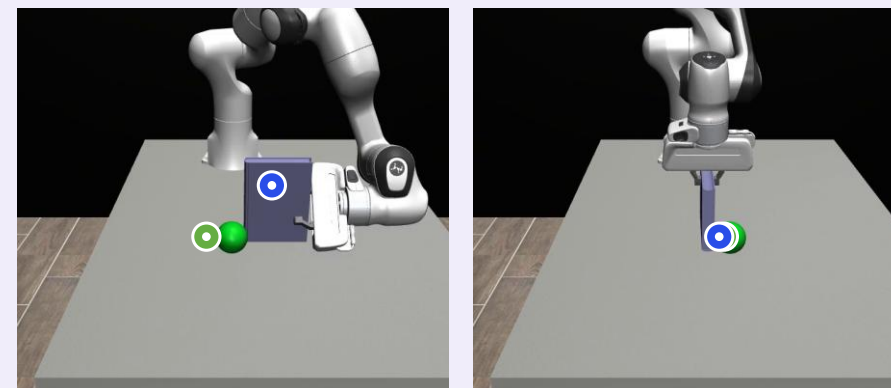
Spine



Spine



Front-top



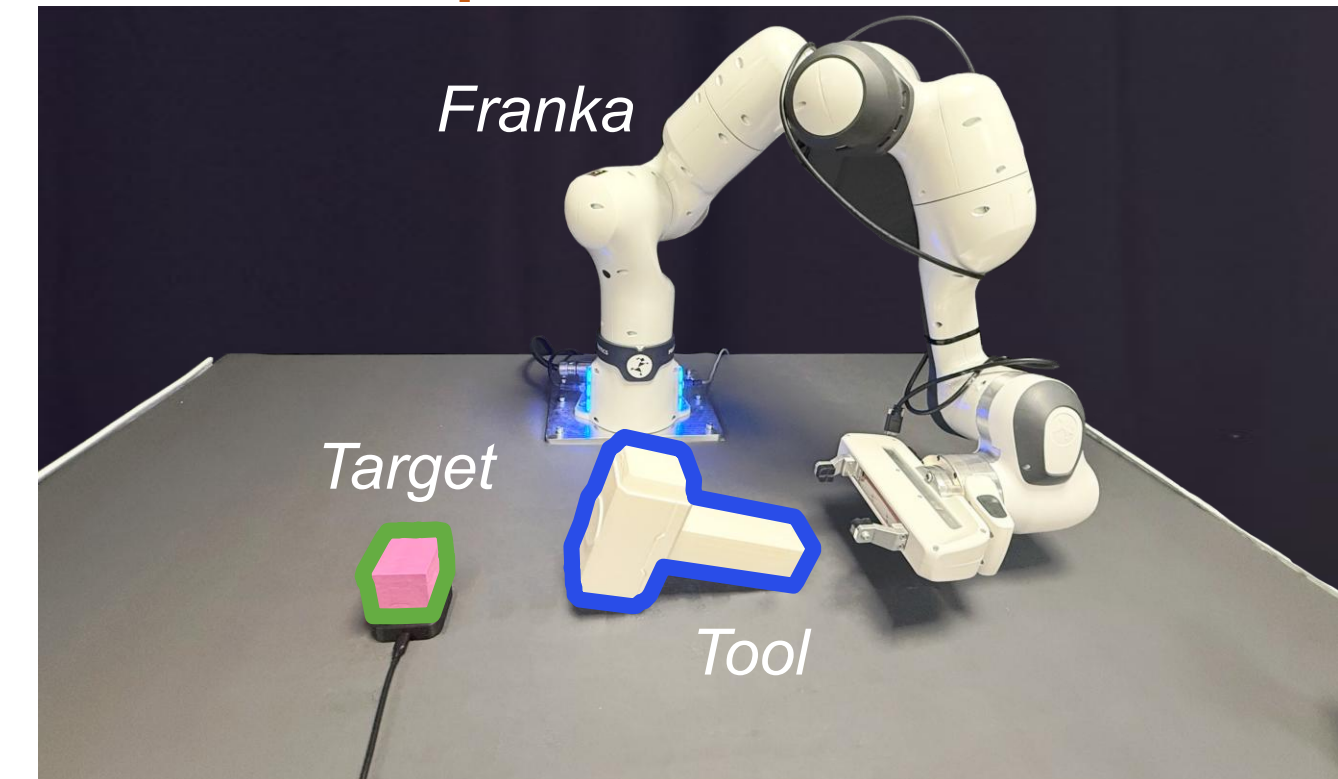
● Target point ● Functional point

(b) Real-world Tasks

7 tools



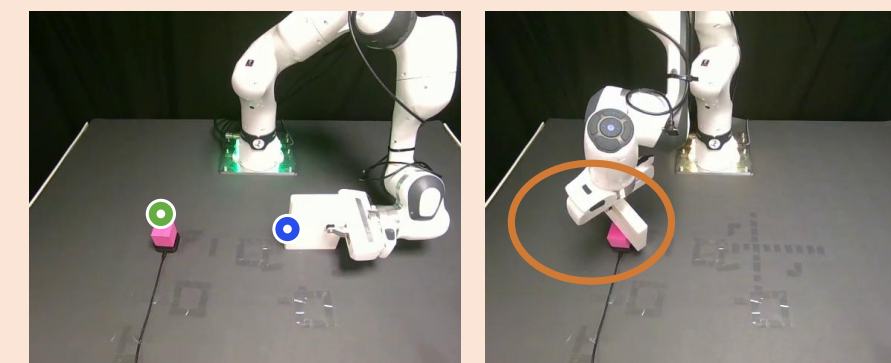
Robot workspace



3 Tasks: Hit / Sweep / Hook with Two Functional Patterns

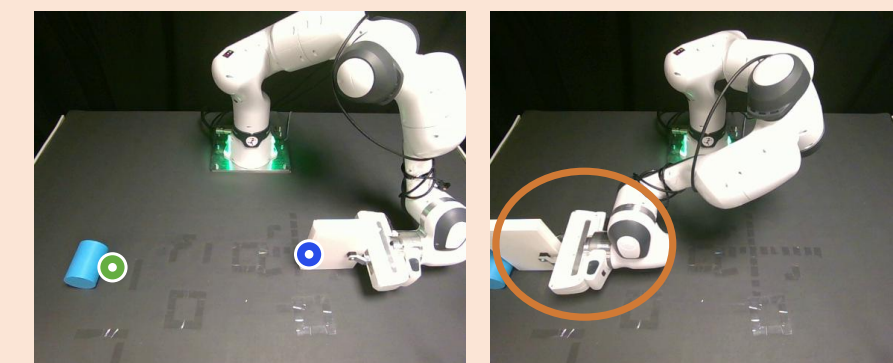
Hitting

Pattern 1



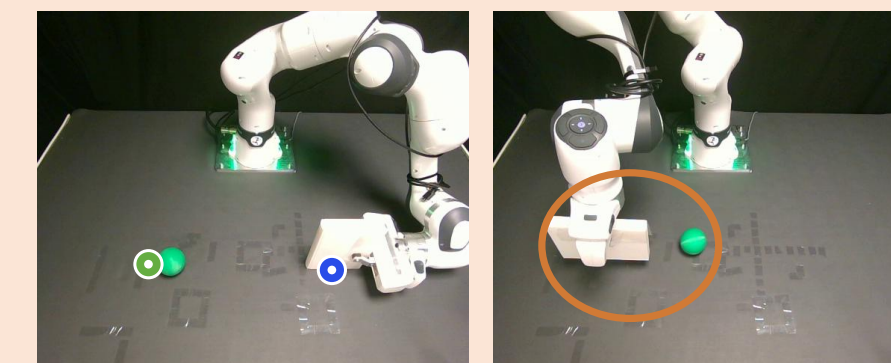
Sweeping

Pattern 1

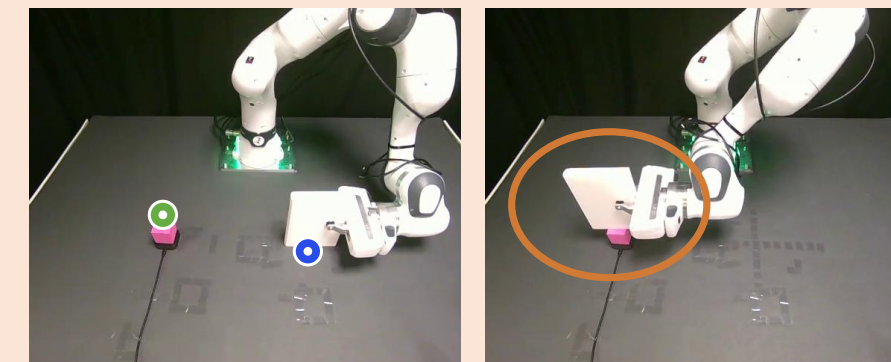


Hooking

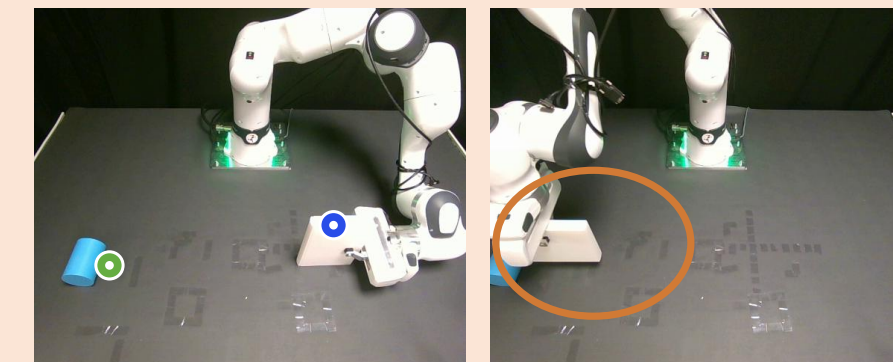
Pattern 1



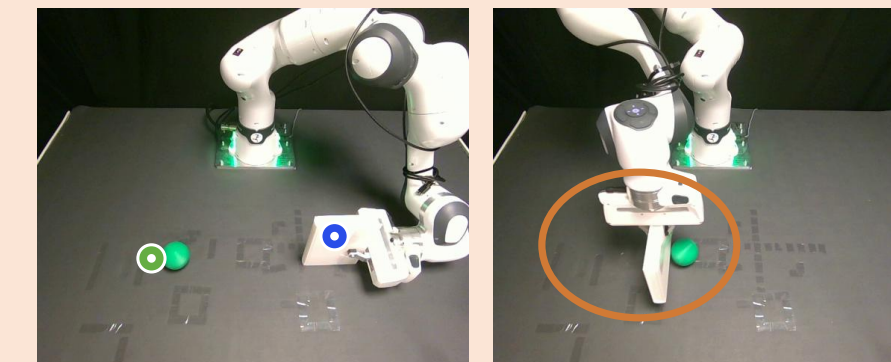
Pattern 2



Pattern 2



Pattern 2



● Target point ● Functional point